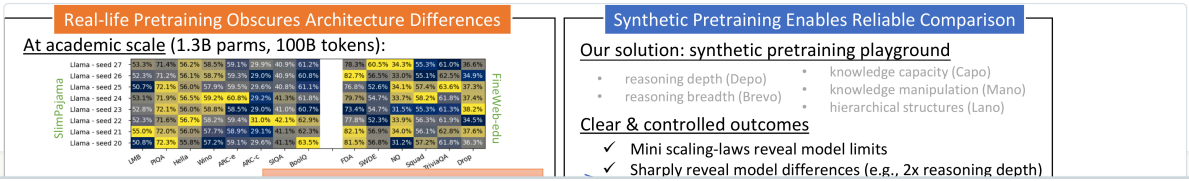
Physics of Language Model

Motivation

Architecture comparison at academic scale is noisy enough to hide real signal. At 1.3B parameters and 100B tokens, random seeds alone can move benchmark accuracy by 2-4 points, often the same size as the reported gain from a new architecture. Pretraining loss is also a weak proxy for reasoning: several linear-state models optimize perplexity well yet underperform on multi-hop tasks. Finally, data quality confounds comparisons because weak corpora suppress capability emergence and blur architectural differences.

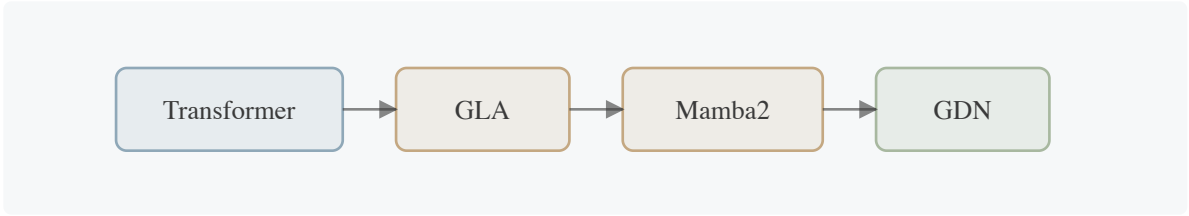
- Challenge 1: loss quality does not equal capability quality.
- Challenge 2: seed noise dominates architecture signal.
- Challenge 3: data and optimization confounds mask causality.



Background

The paper evaluates four architecture families under a shared synthetic setup: Transformer, GLA, Mamba2, and GDN. Canon is introduced as a horizontal residual primitive that can be inserted across families.
Part 4.1 reproduces known qualitative rankings from real-world pretraining but in a cleaner regime: Transformers and GDN generally lead reasoning depth, Mamba2 and GLA favor efficient memory-style behavior, and Canon shifts this frontier by adding explicit local transport between adjacent token states.

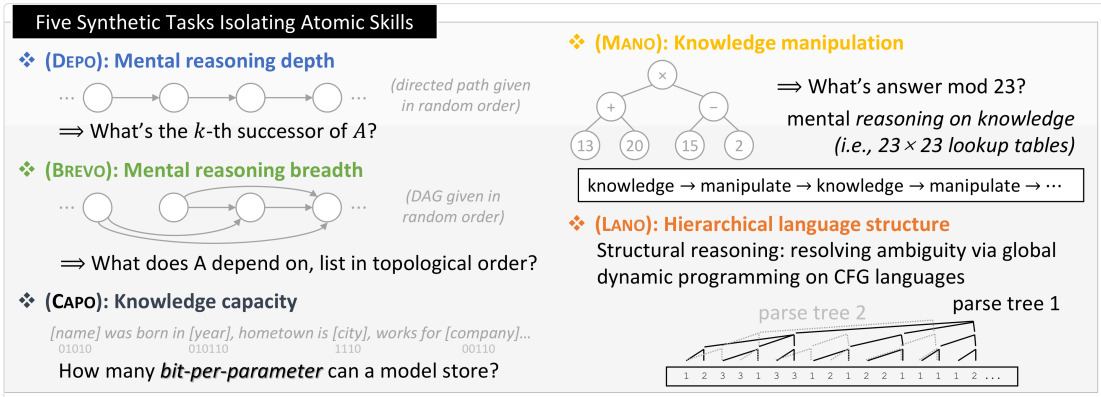
- Transformer: strong reasoning depth, expensive global mixing.
- GLA/Mamba2/GDN: efficient long-context memory via linear-state updates.
- Canon: shared local mixing primitive across both families.



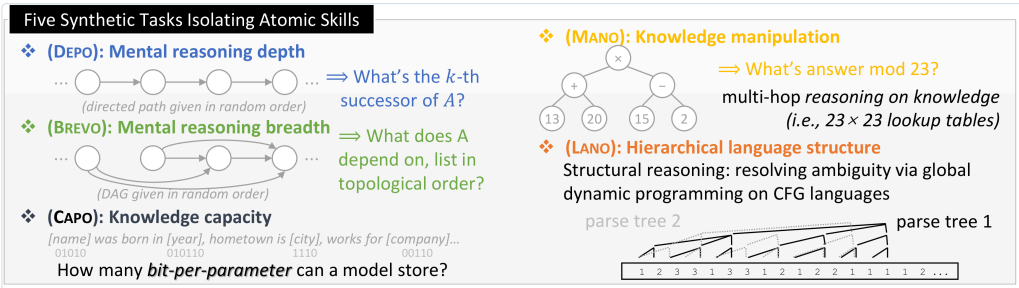
Synthetic Playground

Five pretraining tasks decompose language intelligence into atomic skills: Depo (depth), Brevo (breadth), Capo (capacity), Mano (manipulation), and Lano (hierarchical structure). This removes real-data confounds and enables low-cost architecture science.

- **Depo:** multi-hop graph reasoning with controllable hop count.
- **Brevo:** DAG topological ordering with high branching.
- **Capo:** biological knowledge extraction from synthetic genomes.
- **Mano:** arithmetic chain manipulation and composition.
- **Lano:** context-free grammar understanding and generation.



Five-task design used to isolate architecture behavior.



Key Takeaway

> Synthetic tasks isolate atomic skills, enabling clean architectural comparisons at low cost.

- Low compute budget, high measurement fidelity.
- Controlled depth and breadth reveal causal design effects.
- Findings transfer to trillion-token pretraining.